Muhammad Balawal Safdar

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Skills

Languages: Python, Java, C#, Kotlin, JavaScript, SQL/NoSQL, PL/SQL, Bash
Platforms: Unix/Linux, Git, CI/CD, Docker, Kubernetes, AWS/Azure, JIRA, Confluence, Agile, Postman, AutoCAD
Frameworks/Libraries: ASP.NET Core, FastAPI, Flask, Node.js, Express.js, Next.js, React Native, TensorFlow
ML/AI: TensorFlow, computer vision, LLM APIs, Embeddings

Experience

Software Engineer Intern, **Backboard.io** — **May 2026 – Present**

- Contributed to the development of a **CLI and IDE integration layer** for LLM infrastructure, improving developer workflows for interacting with internal AI tools and APIs.
- Designed and implemented an **LLM anonymization layer** to sanitize sensitive data in prompts, outputs, and logs, enabling safe debugging and evaluation of AI systems.
- Collaborated with **Major League Hacking (MLH)** to support hackathons through volunteer engineering support.

Mobile Applications Developer Intern, **Mogile Technologies (ChargeHub)** — **January 2026 – May 2026**

- Built Android features in **Kotlin** and **Jetpack Compose** for the **ChargeHub** mobile app, including EV station ratings, membership features, and map functionality in an Agile team environment.
- Redesigned the charging session timer system to update locally every second while synchronizing with server data, handling multi-session edge cases, cross-device continuity, and shared **ViewModel** state management.
- Improved engineering productivity by stabilizing **Azure Pipelines** flaky tests, creating custom **GitHub Actions** workflows, and refactoring legacy Java components into idiomatic Kotlin through peer-reviewed pull requests.

CS Tutoring, **John Abbott College** — **March 2025 – December 2025**

- Tutored students in **C# programming**, focusing on OOP and data structures.
- Assisted in debugging, algorithm implementation, test case design, and exam preparation

Projects

PTAgent: Agentic AI Network Automation Tool - CUSEC 2026 Winner

- Designed architecture using **IPC**, **Socket.IO**, and a **FastAPI** backend where an AI agent analyzes network state and executes configuration commands autonomously.
- Built an agentic AI system that automates network topology creation and configuration in **Cisco Packet Tracer**, converting manual workflows into automated tools using **Python** and **JavaScript**.
- Implemented **MLOps-style deployment** patterns for the AI agent with real-time data analysis and automated workflow orchestration.

Sentinel: Traffic Accident Detection System - JacHacks 2025 Winner

- Developed full-stack architecture integrating **MongoDB** (schema design and geospatial queries), **RESTful APIs**, **Leaflet.js** geographic clustering, **Gemini AI** incident reporting, and **Twilio** voice automation.
- Built real-time accident detection system using **TensorFlow COCO-SSD**, a **Next.js** frontend, and web scraping of Quebec 511 camera metadata with bounding box collision detection.

Remoto: Voice-Controlled Remote Computer System - McGill 2025 Winner

- Built a secure voice-controlled remote access system enabling real-time control of a computer from anywhere.
- Developed a custom **AI agent** that analyzes screen state using **OpenCV** and **Tesseract OCR** to generate keyboard and mouse actions via **pyautogui**.
- Implemented a **React** frontend with speech-to-text, **Supabase** (PostgreSQL schema design and authentication), and low-latency live streaming using **FFmpeg**, **MediaMTX**, and **Cloudflare**.

Education

Concordia University, **Bachelor of Computer Science (BCS)** — Present
John Abbott College, **DEC Computer Science** — August 2023 – May 2026

Dean's List recipient

Awards & Achievements

- First Place - JACHacks 2025
- First Place - MariHacks 2026
- Second Place - McGill CodeJam 2025
- Second Place - MariHacks 2025
- Best Agentic AI - CUSEC 2026
- Best Design - MariHacks 2026
- 11labs - JACHacks 2026
- 11labs - MariHacks 2026
- Honorable Mention - DawsHacks 2025